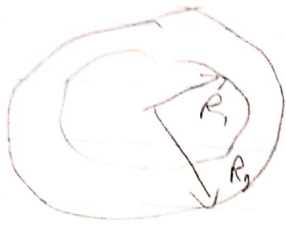


83/100

14.8



$$L = 100 \text{ mm}$$

$$R_1 = 11.3 \text{ mm}$$

$$R_2 = 5.7 \text{ mm}$$

$$3.6 \times 10^{-3} \text{ kg over } 100 \text{ h}$$

$$D = \text{const}$$

$$a) \quad \dot{J}_A = -D_{AB} A \frac{dP}{dr}$$

$$= -D_{AB} (2\pi r l) \frac{dP}{dr}$$

$$\int \frac{J_A}{r} dr = -D_{AB} (2\pi l) dP$$

$$J_A \ln r = -D_{AB} (2\pi l) dP$$

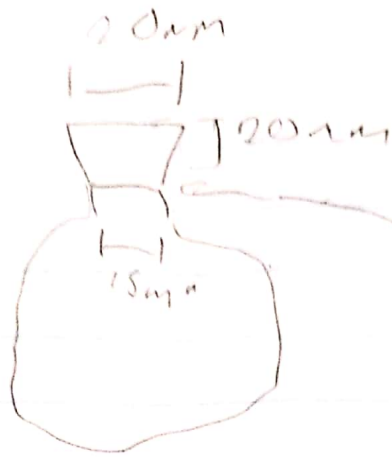
$$J_A = \frac{-D_{AB} (2\pi l) dP}{\ln r}$$

$$\frac{J_A}{-D_{AB} 2\pi l} = \frac{dP}{\ln r} \quad \times$$

b) Since the profile is not linear, the D_{AB} is not constant. D_{AB} increases with carbon concentration.

4.10)

(37/37)



$$2 \times 10^{-3} \text{ kmol/m}^3$$

$$D = 0.15 \times 10^{-9} \text{ m}^2/\text{s}$$

$$J_z = -D_{AB} \frac{dC_A}{dx}$$

$$N_A = -D_{AB} A_c \frac{dC_A}{dx}$$

$$N_A \int_0^L \frac{dx}{A_c} = -D_{AB} \int_{x_{A_1}}^{x_{A_2}} dC_A$$

$$A_c = \pi \frac{R_1^2 + R_2^2}{2}$$

$$= D_{AB} (C_{A_1} - C_{A_2})$$

$$= D_{AB} C_{A_1}$$

$$N_A \int_{R_1}^{R_2} \frac{dR}{\pi R^2} \frac{L}{R_2 - R_1} = D_{AB} C_{A_1}$$

$$\frac{N_A}{\pi} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \frac{L}{R_2 - R_1} = D_{AB} C_{A_1}$$

$$N_A = 3.14 \times 10^{-15} \text{ kmol/s}$$

14.19)

$$N_A A + \frac{d}{dr} (D_{AB} A) \frac{dC_A}{dr} = A \frac{dC_A}{dt} dr$$

33/33

$$A = 4\pi r^2$$

$$N_A (4\pi r^2) + 4\pi \frac{d}{dr} (D_{AB} r^2) \frac{dC_A}{dr} = 4\pi r^2 \frac{dC_A}{dt} dr$$

4\pi r^2

$$N_A + \frac{1}{r^2} \frac{d}{dr} (D_{AB} r^2) \frac{dC_A}{dr} = \frac{dC_A}{dt} dr$$